

Skills Summary

Comparing Unlike Fractions

Use this reference while completing your worksheet or when studying.

Skill 1 — Comparing to the benchmark one half

Benchmark test: a fraction is more than $\frac{1}{2}$ when its numerator is more than half its denominator, and less than $\frac{1}{2}$ when the numerator is less than half the denominator. If one fraction is above the benchmark and the other is below, you are done — no common denominator needed.

Example: Compare $\frac{3}{7}$ and $\frac{1}{2}$.

Step 1. Check the numerator against half the denominator instead of rewriting anything.

Step 2. Half of 7 is 3.5, and $3 < 3.5$, so $\frac{3}{7} < \frac{1}{2}$

Try it: Compare: $\frac{5}{9} \square \frac{1}{2}$

< check

Skill 2 — Comparing with a common denominator

Plan: rewrite both fractions with the same denominator (a number both denominators divide into), then compare the numerators — same-size pieces make the comparison fair.

Example: Compare $\frac{3}{5}$ and $\frac{2}{3}$.

Step 1. Both 5 and 3 divide into 15, so rewrite both fractions in fifteenths.

Step 2. $\frac{3}{5} = \frac{9}{15}$ and $\frac{2}{3} = \frac{10}{15}$, and 9 pieces is fewer than 10, so $\frac{3}{5} < \frac{2}{3}$

Try it: Rewrite in twelfths, then compare: $\frac{3}{4} \square \frac{5}{6}$

> check

Skill 3 — Making equivalent fractions

Rule: multiplying the numerator and denominator by the same number makes an equivalent fraction — the same amount cut into smaller pieces. This is the move that builds every common denominator.

Example: Fill in: $\frac{3}{4} = \frac{\square}{8}$

Step 1. Ask what number times 4 gives 8, then multiply the top by the same number.

Step 2. $4 \times 2 = 8$, so the numerator is $3 \times 2 = 6$

Try it: Fill in: $\frac{4}{5} = \frac{\square}{15}$

12 check

(!) Watch out: you cannot compare unlike fractions by looking at numerators alone or denominators alone. Always compare same-size pieces (a common denominator) or use a benchmark.